

A Fast Wake-Up Power Gating Technique with Inducing a Balanced Rush Current

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Abstract—Power gating technique is important for saving the leakage power, especially for the very deep-submicron system-on-a-chip designs. Although the multi-threshold voltage CMOS (MTCMOS) technology can enable us to cut down the leaky path easily, the induced power/ground bounce is getting worse and can not wake up from power down mode quickly. In this paper, we propose a new concept of power gating technique by balancing the variations of rush current to accelerate the wake-up procedure for a given power/ground bounce specification. Performance evaluations of a 40-bit ALU circuit using the TSMC 0.18 μ m CMOS technology show that our proposed power gating technique can achieve a 10.23% reduction in wake-up time while keeping the power bounce specification, as compared with the conventional power gating technique.

I. INTRODUCTION

Thanks to the advancing semiconductor technology, more than millions of transistors can be easily integrated into single silicon chip nowadays. It inevitably consumes lots of power at the same time, thus the designing for low power emerges to be the design trend of integrated circuit (IC) development. The power consumption for an integrated circuit is mainly composed of dynamic power and static power. Integrated circuit consumes the so-called dynamic power in the case of charging and then discharging the capacitive load to evaluate the desiring status while the static power is consumed in the form of direct current (dc) and leakage current during standby or sleep mode. Equation (1) represents the dynamic power consumption, where C_L , V_{dd} , and f are the load capacitance, the supply voltage and the frequency of transitions, respectively.

$$P_{dynamic} = C_L \times V_{dd}^2 \times f \quad (1)$$

Therefore, reducing the supply voltage is effective to reduce the dynamic power in a quadratic ratio. Over the past decades, the supply voltage has been decreased from 5V to sub-1V as the scaling down of the semiconductor technology. At the same time, we should also apply the device with lower threshold voltage so as to keep the circuit performance as well. As a result, the increasing of leakage current will also increase the static power consumption. For the case of system-on-a-chip (SOC) design with integrating more than millions of

gates, many individual functional blocks (e.g. processor and peripherals) may be partitioned into different supply voltage domains while several operating modes are designated such as active mode, standby mode, sleep mode, and so on. In the standby mode or sleep mode, the system chip may have some functional blocks being activated and have others being disabled or even shut down all functional blocks. In these cases, existing of large leakage current will seriously worsen the overall power consumption. One method to solve this problem is to apply a power transistor with high threshold voltage to isolate the power supply and the internal circuits which are operating in the shut down status. This technique is so-called power gating [1][2].

In many battery supplied and power managed applications such as cell phone and pad, wake-up period of power gating is an important specification. Power gating with fast wake-up mechanism can provide the system a longer time period to stay in sleep mode so as to save more static power. As shown in Fig. 1, for each group of data to be processed, the shorter the wake-up period is, the longer the sleep time can be. Furthermore, for those system chips which are manufactured using very deep sub-micron process technology (ex: 65nm, 45nm or below), power gating with fast and cycle-based wake-up is emerging to power down most unused functional blocks as long as possible to save more leakage power.

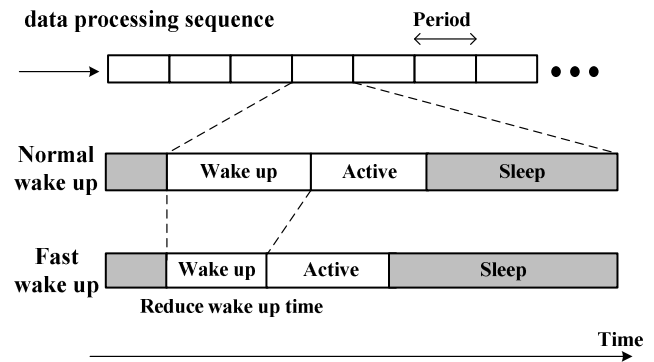


Figure 1. The importance of wake-up time in power gating techniques.

However, to wake up fast from the power down mode will generate great power/ground bounce inevitably because of the instantaneously generated rush current. The power/ground noise will then disturb the operation of the activated functional blocks. There are many researches devoted to solve this problem [3-7] with paying many design efforts. In this paper, we propose a new power gating technique by balancing the variations of wake-up rush current, which can shorten the wake-up time while minimizing the power/ground bounce.

The rest of this paper is organized as follows. We will briefly review the power gating techniques in Section II. Section III will focus on the proposed power gating technique with inducing a balanced rush current. Section IV describes the test circuit and performance evaluations. Finally, we conclude this paper in Section V.

II. OVERVIEWS OF THE POWER GATING TECHNIQUES

Fig. 2(a) illustrates the circuit structure of power gating technique with including the equivalent model of the package where the power transistors ($M_{p,a}$ and $M_{p,b}$) are high threshold voltage devices and PCB_vdd and PCB_gnd represent the supply source and ground of the system board, respectively. When the “Circuit A” enters the sleep (power down) mode, the sleep control signal ($sleep_a$) will become high so that the high leakage current path from PCB_vdd to PCB_gnd is limited due to the high threshold voltage power transistor ($M_{p,a}$). When the “Circuit A” enters the active (normal function) mode, the sleep control signal ($sleep_a$) transits from high to low gradually and turn on the power gating transistor gradually. After the virtual power supply ($vvdd_a$) reaches the value of 90% of supply voltage (PCB_vdd) and begin the active mode, the “Circuit A” can start to operate normally. The time period between the end of sleep mode and start of active mode is defined as the wake-up time, as shown in Fig. 2(b). In this period, a large rush current (I_{rush}) is emerging abruptly from PCB_vdd to charge both the capacitances of the internal nodes ($C_{int,a}$) and the virtual supply ($C_{vdd,a}$). The varying rush current flows through the inductance of the bonding wire and generates the so-called power bounce on the on-die supply voltage (die_vdd), as shown in Fig. 2(c). The induced power bounce may disrupt the operations of the adjacent circuits, e.g. “Circuit B” of Fig. 2(a). Obviously, a short wake-up time requires a larger I_{rush} and therefore results in a serious power supply variations on the adjacent circuits which operate in active mode. In worst case, the circuit which operates in the active mode may violate the time specification and work abnormally.

Since there is a trade-off relation between the power/ground bounce and the wake-up time, many papers devoted to improve one of these two parameters in the last few years. We will categorize and describe them briefly below for reference.

A. Small Power/Ground Bounce [3][4][5]

During the wake-up period, the voltage level of power/ground bounce is mainly determined by the capacitive load of the virtual supply voltage. Large capacitance requires a large rush current for a given wake-up time specification. Ref [3] propose a capacitance reduction method to reduce the

power/ground bounce, and [4] controls the substrate voltage of power gating transistor to limit the power/ground bounce. Ref [6] proposed a non-uniformed multi-step power gating mechanism to reduce the instantaneously peak current.

B. Fast Wake-Up Time [6][7]

Both [6] and [7] use two separated power line architecture, one is the main power line and another is the bypass power line. Main power line supplies the power source in the active mode while the bypass power line provides the power source during the wake-up period. Therefore, the required rush current will induce the power/ground bounce on the bypass power line rather than on the main power line so that the induced disturbance on power supply is isolated from the circuit which is operating in the active mode.

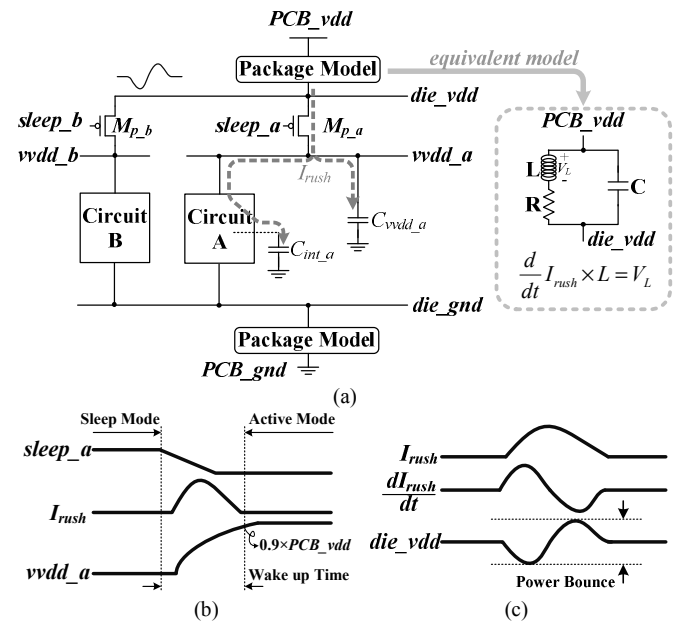


Figure 2. (a) The circuit structure of the conventional power gating technique [1], (b) the conceptual waveforms of the rush current, and (c) the conceptual waveforms of the power/ground bounce.

Either complicated controlling mechanism [4] or extra power line routing resource [7] is required to improve the wake-up time or reduce power/ground source. We propose a simple architecture of power gating, basing on the most original power gating technique [1] to significantly reduce the wake-up time while generating small power/ground bounce.

III. THE PROPOSED POWER GATING TECHNIQUE

In this section, a newly proposed power gating technique which reduce the power/ground bounce and achieve fast wake-up by balancing the variations of the required rush current. The design concepts are introduced first and then the key circuit implementations are shown.

A. The Design Concepts

From Fig. 2(c), the power bounce is resulted from instant change of rush current. Thus we propose the method to induce the rush current in a more balanced way by dividing the single power gating transistor ($M_{p,a}$) of the conventional design [1]

into two groups. The resulting power transistor structures are re-drawn in Fig. 3 where the power transistor of the conventional design (M_{p_a}) is divided into M_{p_a1} and M_{p_a2} and the channel width summation of transistors M_{p_a1} and M_{p_a2} equals to M_{p_a} . Fig. 3 also shows the conceptual waveforms of the related power gating control signals and the induced rush current of both the conventional power gating technique ($sleep_a$ and I_{rush1}) [1] and the proposed power gating technique ($sleep_a1/sleep_a2$ and I_{rusha1}/I_{rusha2}). One of the control signals of the proposed power gating technique ($sleep_a1$) operates the same as the conventional power gating technique while the other one ($sleep_a2$) operates in a two-step waveform which is similar to [5]. The interfered on-die power supply signal (die_vdd) is also shown in Fig. 3.

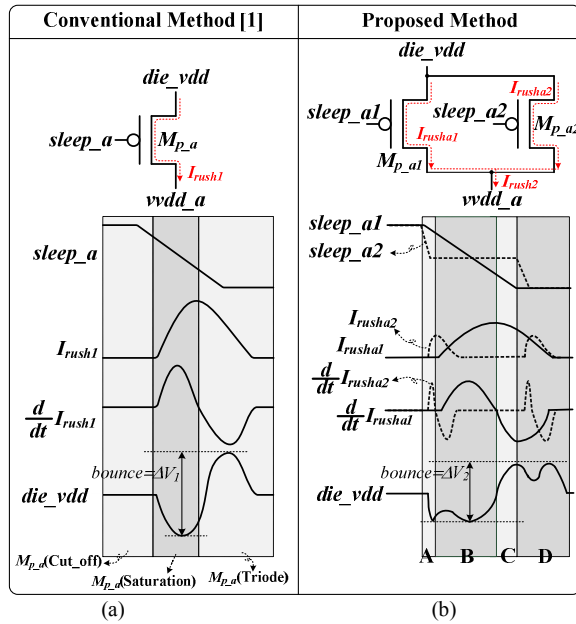


Figure 3. (a) The conventional power gating technique [1], and (b) the proposed power gating technique with balancing rush current.

Signals $sleep_a1$ and $sleep_a2$ partition the wake-up period into four phases which are shown as A, B, C, and D in Fig. 3(b). In phase “A”, transistors M_{p_a1} is turned off and M_{p_a2} is switched from cut-off region to saturation region. Because of a smaller transistor size of M_{p_a2} , the variation of the induced rush current ($dI_{rusha1}/dt + dI_{rusha2}/dt$) is smaller than that of the conventional power gating technique (dI_{rush1}/dt) so that a smaller power bounce is generated. After signal $sleep_a2$ stops decreasing at a medium value between supply voltage and ground level to turn M_{p_a2} on a constant strength, the proposed power gating technique enters phase “B”. In this period, the dI_{rusha2}/dt generated by M_{p_a2} is zero to contribute no power bounce on die_vdd while M_{p_a1} works similar to M_{p_a} but with generating a smaller power bounce because dI_{rusha1}/dt is smaller than dI_{rush1}/dt which is generated by a bigger transistor M_{p_a} . As the virtual supply voltage ($vvdd_a$) increases, I_{rush2} ($I_{rusha1} + I_{rusha2}$) starts to decrease as well. The power bounce goes upward and drive die_vdd to exceed the level of supply voltage, now the circuit enters the third phase “C”. Similar to the operation of phase “B”, the induced power bounce of our proposed power gating is all generated by M_{p_a1} and is still smaller than that generated by M_{p_a} . Finally,

$sleep_a1$ becomes low and $sleep_a2$ starts to decrease to low, the circuit enters phase “D”. In this period, I_{rusha1} decrease as $vvdd_a$ increases to supply voltage level while I_{rusha2} increases first and then decreases to zero. The combinations of dI_{rusha1}/dt and dI_{rusha2}/dt drive die_vdd to increase first and then decrease to stable at the supply voltage level, as shown in phase “D” of Fig. 3(b).

The proposed power gating technique which applies two individually controlled power transistors generates a small variation on rush current by balancing the instantaneous peak current in the whole wake-up period, and the result is a small power bounce ($\Delta V_2 < \Delta V_1$ in Fig. 3).

B. The Circuit Implementation

A simple controller is modified from [5] to generate the required signals illustrated in Fig. 3(b). The resulted circuit and the corresponding signal waveforms are shown in Fig. 4 and Fig. 5, respectively. The power gating controller receives the original one-step sleep control signal ($sleep_a$) and then generates two required sleep control signals ($sleep_a1$ and $sleep_a2$). The controller is powered by die_vdd so as to operate normally during the wake-up period. In the initial condition, $sleep_a1$ and $sleep_a2$ all stay at high level. When signal $sleep_a$ changes to low, first the slew rate adjustable buffer will slow down the transition of $sleep_a$ and generate a more gradual signal $sleep_a1$. Then the delay element and an XNOR gate are used to produce a return to one pulse on node $V1$ and turn transistors M1 and M4 on in a short while. When node $V1$ becomes low, node $V2$ is getting charged through the voltage divider formed by transistors M2/M3 and then bias $V2$ at a medium voltage level. The medium voltage level can be any arbitrary value between supply voltage and ground depending on the transistor dimension of M2 and M3. Then signal $sleep_a2$ will be discharged through M4 to the medium voltage level. Note that M1 and M4 are turned on only in a short time, the dc current resulted by M2 and M3 to generate $V2$ can be small enough to be ignored. After $V1$ becomes high and $sleep_a$ keeps low, transistor M6 is turned on to discharge $sleep_a2$ to the ground.

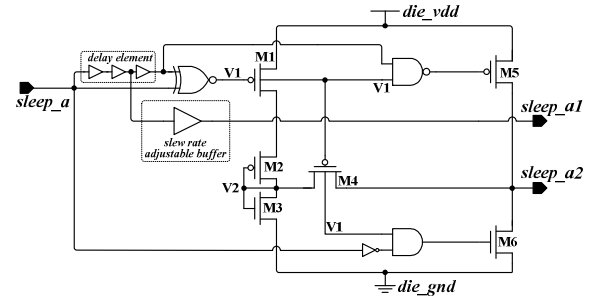


Figure 4. The controller circuit used to create the sleep control signals.

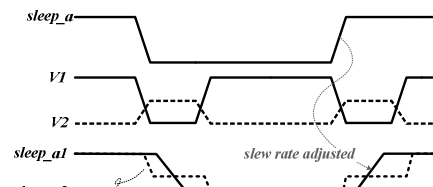


Figure 5. The conceptual waveforms of the sleep control signals in Fig. 4.

IV. PERFORMANCE EVALUATIONS

A. The Test Circuit

In a high performance microprocessor design using a very deep-submicron technology, the power management unit will power down and wake up some functional blocks in several clock cycles so as to cut down any possible leakage current. Arithmetic logic unit (ALU) which consists of combination logic for logic operations, arithmetic circuit for arithmetic operations and sequential elements to be the temporary registers, is a key and complex enough functional block of a microprocessor. Therefore, we use the ALU as the test circuit to verify our proposed power gating technique and the test chip architecture is shown in Fig. 6. In Fig. 6, we use two linear feedback shift register (LFSR) to generate two 40 bits pseudo random input patterns and output the results serially by the left-shift function of ALU for saving I/O pins. Signals *rst* and *In_en* are used to clear and enable the LFSR while *op_code* is used to indicate the operations of ALU. Two power gating controlling signals *sleep_a1* and *sleep_a2* are generated by the power gating controller shown in Fig. 4. The test chip is designated using 1.8V TSMC 0.18 μ m CMOS process technology and the chip layout is also shown in Fig. 6.

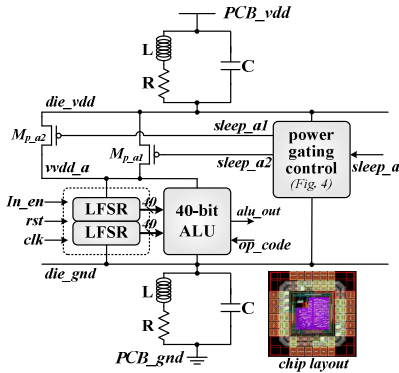


Figure 6. The architecture and whole chip layout of the test chip.

B. Simulation Conditions and Results

The test circuit with applying both the conventional power gating technique [1] and the proposed power gating technique are designed and simulated using 1.8V TSMC 0.18 μ m MTCMOS process technology where the power transistor is applying the high threshold voltage device and the ALU circuit is applying the low threshold voltage device, the post-layout simulation results are shown in Fig. 7. In the simulations, we define a power bounce specification first, to be less than $\pm 5\%$ of the supply voltage, and then measure the wake-up time. From Fig. 7, we can prove that the design with applying our proposed power gating technique can achieve a 10.23% (2.72ns vs. 3.03ns) reduction on wake-up time while keeping the same power bounce specification (about $\pm 0.83V$), as compared to the conventional power gating technique [1]. The simulation results are also summarized in Table I.

TABLE I. SUMMARY OF SIMULATION RESULTS

Power Gating	Power Bounce (V)	Wake-Up Time	Reduction
Convention [1]	(0.084+0.083)	3.03 ns	-
Our Proposed	(0.083+0.083)	2.72 ns	10.23%

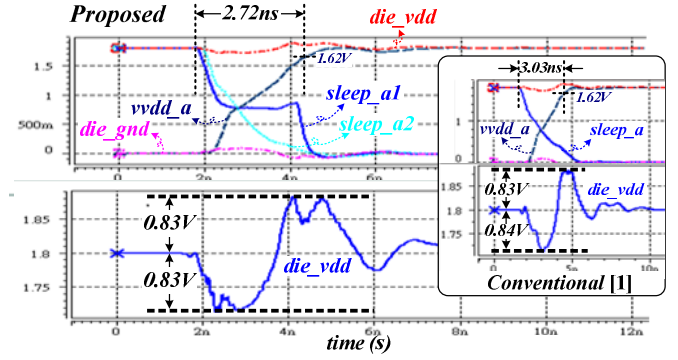


Figure 7. The post-layout simulation waveforms of the ALU circuit with applying our proposed and the conventional power gating techniques.

V. CONCLUSION

Power gating technique with the feature of cycle-based wake up is contributive to save the leakage power as many as possible because all the unused functional blocks can stay in power down mode as long as possible. The key to achieve the cycle-based wake-up is the fast wake-up mechanism with inducing small power/ground bounce. The test circuit using our proposed power gating technique is shown to achieve 10.23% less wake-up time with keeping the same power bounce, as compared with the test circuit using the conventional power gating technique.

ACKNOWLEDGMENT

The authors would like to thank the National Chip Implementation Center (CIC) of Taiwan for providing CAD tools and the chip implementation service.

REFERENCES

- [1] S. Mutoh, T. Douseki, Y. Matsuya, T. Aoki, S. Shigematsu, and J. Yamada, "1-V power supply high-speed digital circuit technology with multithreshold-voltage CMOS," IEEE J. Solid-State Circuits, vol. 30, no. 8, pp. 847-854, August 1995.
- [2] H. Kawaguchi, K. Nose, and T. Sakurai, "A super cut-off CMOS (SCCMOS) scheme for 0.5-V supply voltage with picoampere stand-by current," IEEE J. Solid-State Circuits, vol. 35, no. 10, pp. 1498-1501, October 2000.
- [3] Suhwan Kim, Chang Jun Choi, Deog-Kyoon Jeong, Stephen V. Kosonocky and Sung Bae Park, "Reducing Ground-Bounce Noise and Stabilizing the Data-Retention Voltage of Power-Gating Structures," IEEE Trans. Electron Device, vol. 55, no. 1, pp. 197-205, January 2008.
- [4] Chung-Yu Chang, Wei-Ben Yang, Ching-Ji Huang, and Cheng-Hsing Chien, "New Power Gating Structure with Low Voltage Fluctuations by Bulk Controller in Transition Mode," IEEE Int. Symps. Circuits and Systems, no. 2, pp. 3740-3743, 2007.
- [5] Suhwan Kim et al., "Minimizing inductive noise in system-on-a-chip with multiple power gating structures," in Proc. 29th European Solid-State Circuits Conf., pp. 635-638, 2003.
- [6] Ken-ichi Kawasaki, Tetsuyoshi Shiota, Koichi Nakayama and Atsuki Inoue, "Experimental measurement of a novel power gating structure with intermediate power saving mode," Int. Symps. Low Power Electronics and Design, no. 2, pp. 146-147, 2008.
- [7] Ken-ichi Kawasaki, Tetsuyoshi Shiota, Koichi Nakayama and Atsuki Inoue, "A Sub- μ s wake-up time power gating technique with bypass power line for rush current support," IEEE J. Solid-State Circuits, vol. 44, no. 4, pp. 1178-1183, April 2009.